

Resolvent solve at $L = 0.3$:

R is not coercive, shift $s = 0.05$ works on V_n ,

operator R -gap open

PRIME.RDAGGER.RESOLVENT_SOLVE.01 — round 488

Stefan Hamann

August 31, 2026

Abstract

Route (a) of r486: invert $R = A^+ + \Pi$ on the certified enemy modes of K_{t_c} . **Coercivity first.** $\lambda_{\min}(R)$ on the Nyström space collapses with n : 0.337 ($n = 48, m = 8$), 0.106 ($n = 96, m = 20$), 0.076 ($n = 128, m = 16$). A^+ vanishes on $|t| < t_c$; Π lifts one direction. R is positive semi-definite but not coercive. Shift $R_s = R + s$, $A_s^- = A^- + s$ with $s = 0.05$ is mandatory; $R_s \succeq sI$ as forms. The Birman–Schwinger inequality is shift-invariant ($Q_W \succeq 0$ iff $\text{BS} \leq 1$ for every s); s only converts residual to inverse error, $\|u - R_s^{-1}\varphi\| \leq r/s$. $s \rightarrow 0$ recovers r486 (0.993651). At $s = 0.05$ the correct G -certificate $\lambda_{\max}(G^{1/2}BG^{1/2})$ with $G = P_ER_s^{-1}P_E$ is 0.99415 ($N = 3$) and 0.99439 ($N = 5$). Discrete residual in V_n is $2 \cdot 10^{-16}$. $N = 3$ plus $|\sigma_A(0)| \cdot \text{rest}/s$ overshoots (1.078) and is not a SATZ pad. $N = 5$ tail $5.5 \cdot 10^{-7}$ is negligible. The discrete G is the inverse of the $T = 150$ Galerkin R , not of the operator R on L^2 . Verdict REDUCED(R-not-coercive; s=0.05-G-cert-N5-BS=0.994; operator-R-gap-open). Fourier–HS is not reused. Yoshida–Bombieri zone. No RH claim.

Status. REDUCED(R-not-coercive; s=0.05-G-cert-N5-BS=0.994; operator-R-gap-open). No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. R is not coercive; the shifted G -certificate is < 1 on V_n ; the operator R -gap is open. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$ (scramble-sensitive, vacuous). $R = A^+ + \Pi$, $A^- = |\sigma_A|1_{|t| < t_c}$, t_c as in r479. The r469/r476–r486 contract is in force.

2 Coercivity of R (first)

Nyström (n, m)	$\lambda_{\min}(R)$	$\lambda_{\min}(R_{\text{even}})$
(48, 8)	0.337	0.625
(48, 16)	0.336	0.623
(96, 20)	0.106	0.297
(128, 16)	0.076	0.186

The infimum of the form R on $L^2[-L, L]$ is 0. A shift $s > 0$ is required for a uniform coercivity constant. We take $s = 0.05$.

3 G -certificate

Solve $R_s u_i = \varphi_i$ in the Slepian–Galerkin space V_n . Residual $\leq 2 \cdot 10^{-16}$, so $\|u_i - R_s^{-1} \varphi_i\| \leq r/s$ is vacuous on V_n . The 3×3 block of $G = P_E R_s^{-1} P_E$ at $s = 0.05$, $n = 96$, $N = 5$ is

$$G_{012} \approx \begin{pmatrix} 0.844 & 1.2 \cdot 10^{-4} & 0.303 \\ 1.2 \cdot 10^{-4} & 2.437 & 9.0 \cdot 10^{-3} \\ 0.303 & 9.0 \cdot 10^{-3} & 1.263 \end{pmatrix}.$$

The Birman–Schwinger number is $\lambda_{\max}(G^{1/2} B G^{1/2})$, $B = A^-|_E + sI$, *not* $\lambda_{\max}(BG)$ (the asymmetric product reads 1.023 and is anti-listed).

Object	$N = 3$	$N = 5$	comment
$s = 0$ Galerkin (r486)	0.993651	—	midpoint hit
$s = 0.05$ $G^{1/2} B G^{1/2}$	0.99415	0.99439	margin $5.6 \cdot 10^{-3}$
$+ \sigma(0) \text{rest}/s$	1.078	0.99440	$N = 3$ kills
Higham μ of $Q _E$	$2.87 \cdot 10^{-3}$	$2.33 \cdot 10^{-3}$	finite PD

Lemma 1 (shifted G -certificate on V_n). *At $s = 0.05$, $N = 5$, $T = 150$, $\lambda_{\max}(G^{1/2} B G^{1/2}) + |\sigma_A(0)| \cdot \text{rest}/s \leq 0.99440 < 1$ on the $n = 96$ Nyström space. The $N = 3$ crude tail overshoots and is not used.*

$T = 400$ is not used.

4 Why this is not λ_*

The discrete G inverts the Galerkin R at $T = 150$, not the operator R on $L^2[-L, L]$. Omitting $[T, \infty)$ makes R smaller and the Birman–Schwinger number larger (conservative in sign) only if the $[t_c, T]$ quadrature does not overestimate A^+ . That Filon/complement gap is the r483–r485 leftover, now isolated as an operator–Galerkin difference for R_s^{-1} on five functions. No δ for the operator statement is claimed.

5 Kills

Object	Round	Status
Fourier–HS / K_8 -garbage / $20\times$	r478–r484	not reused
Flat $c_{\text{tail}}I$ as M_∞	r485	undershoot
Unshifted R^{-1} as coercive	r488	$\lambda_{\min}(R) \rightarrow 0$
$N = 3 + \sigma(0) \text{rest}/s$ as SATZ	r488	$1.078 > 1$
$\lambda_{\max}(BG)$ as BS	r488	not the quadratic form
$T = 400$ quadrature	r486/r488	Dirichlet Q jumps
Discrete G -certificate as SATZ	r488	Galerkin only

6 Scaling at $L = 0.8$

The enemy becomes the negative part of $\sigma_A - \sigma_P$, still band-limited plus finitely many cosine nodes; the enemy dimension stays small. P is live. The coercivity of $A^+ + \Pi$ does not improve: a shift remains necessary. The operator–Galerkin gap for R_s^{-1} does not become easier.

7 What is not claimed

No $\lambda_*(0.3) \geq c$. R is not coercive; the shifted G -certificate is < 1 on V_n ; the operator R -gap is open. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.